ROADSAATHI · TOWING & ROADSIDE ASSISTANCE PLATFORM

Operations & Compliance Handbook

Dispatch, partners, support and grievances, damage claims, finance, admin panel, incident response and legal map

Audience	For operations, finance and compliance
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> Confidential. Third-party figures are cited; RoadSaathi projections are planning assumptions. Legal and tax points must be confirmed by qualified advisers.

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Operations standard operating procedures (SOPs)

For the 24×7 dispatch and support desk, partner team and finance. Admin screens are described in `../ADMIN-GUIDE.md`.

1. Desk structure

- **Shifts:** 06:00–14:00, 14:00–22:00, 22:00–06:00, with at least one dispatcher per shift and an on-call ops lead.
- **Tools:** Admin → Dispatch board (always open), Bookings, Partners, Notification log; desk phone; WhatsApp business.
- **Handover:** at the end of each shift, note open jobs, partners offline unexpectedly, and unresolved complaints.

2. Dispatch SOP

Situation	Action	Target
New booking, offers going out	Watch only; system offers N nearest drivers per round	Assigned < 3 min
No acceptance after round 1	Call the nearest online partner directly; ask them to accept in the app	—
`NO_DRIVER_FOUND`	Call partners in neighbouring zones; **manual assign** in the booking; call the customer with an honest ETA	Customer called < 5 min
Driver not moving 5 min after assignment	Call driver; if unreachable, reassign (status back to finding driver)	—
ETA slipping > 15 min	Proactive call or SMS to the customer	—
Accident recovery booking	Ask: is anyone injured (direct to **112**)? Is there a police case/FIR? Do police need to release the vehicle? Record answers in booking notes before dispatching	Before dispatch
Lockout	Driver must check proof of ownership (RC and ID) before opening	At site
Unsafe location (highway shoulder at night, flood)	Advise the customer to move to a safe spot and stay out of the vehicle; prioritise; brief the driver	—
Scheduled booking	Released automatically before the slot; confirm with the customer 30 min before	—

3. Partner onboarding and quality

1. **Application review (within 24 h):** check the licence class, RC (body type: recovery vehicle or crane), insurance, permit, fitness and PUC validity, PAN/GSTIN, bank/UPI, and police verification for drivers.
2. **Physical check** for the first truck of each partner: equipment (straps, wheel dollies, winch, cones, reflective jacket), photos of the truck.
3. **Approve with commission** → agreement is generated and emailed → partner e-signs → **active**.
4. **Training (30 min):** app flow, trip PIN, handling standards (flatbed for luxury, low, AWD, EV and damaged vehicles), photos at pickup and drop, conduct, no off-platform solicitation, cash handling and commission remittance.
5. **Quality rules:** rating below 4.0 over 20 jobs → coaching; below 3.5 or a serious complaint → suspension pending review; any safety incident → immediate suspension.
6. **Document expiry:** review monthly. Expired insurance or fitness → partner set offline until renewed.

4. Customer support and grievance SOP

- **Channels:** phone, email (support address), WhatsApp, contact form (Admin → Leads).
- **Legal timelines (E-Commerce Rules): acknowledge every complaint within 48 hours** and **resolve within one month**. Our internal target is resolution within 7 days.
- **Register every complaint** (until the grievance register ships, use Admin → Leads with status, plus a shared sheet) with booking code, category, owner, received/acknowledged/resolved timestamps and outcome.
- **Categories and owners:** billing/refund → finance; damage → ops lead; driver conduct → partner team; privacy/data request → founder (privacy); app bug → engineering.
- **Escalation:** anything unresolved after 7 days goes to the Grievance Officer.

5. Damage-claim SOP

1. Customer reports within **24 hours** with photos (per Terms). Open a claim with the booking code.
2. Collect evidence: pickup and drop photos (once photo capture ships), booking timeline and GPS, driver statement.
3. Decide within 7 days: pre-existing / partner liable / shared. Partner pays directly, or through their insurance, or by recovery from settlements as the partner agreement allows (clause 5.2).
4. Pay or reject in writing with reasons. Log every claim amount for the risk register (R5).

6. Finance SOP

- **Daily:** reconcile Razorpay settlements (when live) with `Payment` rows.
- **Weekly settlement:** Admin → Payouts & ledger; pay partners their net for online jobs; collect commission + GST for cash jobs (offset where possible); mark settled with UTR/reference.
- **Credit control:** a partner owing more than the set limit or for longer than 14 days is paused until cleared.
- **Monthly:** export invoices for GST returns; TDS/TCS per the model chosen with the CA; month-end close; bank reconciliation.
- **Refunds:** per the Cancellation & Refund policy (5–7 working days to original method); record the reason.

7. Data handling and retention

- Look up customer data only to serve that customer's booking or complaint.
- Never share a customer's phone, address or vehicle details outside the assigned partner and staff.
- Data requests (access, correction, deletion) go to the privacy owner and are answered within the time the DPDP Rules set (once in force). Log each request.
- Proposed retention: bookings, invoices and ledger as required for tax records (CA to confirm, typically 8 years); notification and system logs 180 days; inactive customer accounts reviewed after 3 years.

8. Daily checks (ops lead)

- [ ] Partners online per zone for tonight
- [ ] Yesterday's `NO_DRIVER_FOUND` and cancellations, with reasons
- [ ] Complaints near the 48-hour acknowledgement deadline
- [ ] Partner balances owed > 14 days
- [ ] Notification log failures (SMS/email)

Admin guide — what you can change without touching code

Sign in at `/en/admin` with the `ADMIN_EMAIL` address (emailed one-time code). The super admin can add **Dispatchers** (operations only) and **Admins** under *Users & staff*.

I want to…	Where
Rename the brand, change logo/colors, tagline, social links	Settings → **Brand** (updates site, emails, invoices, SMS, contracts, favicon, PWA name instantly)
Change phone numbers, emergency number, email, office address, hours	Settings → **Contact & address**
Set GSTIN, PAN, SAC code, GST %, invoice/booking prefixes	Settings → **Business, GST & invoicing**
Change night/surge multipliers, cancellation fee, default commission	Settings → **Pricing rules**
Tune dispatch radius, offers per round, timeouts, scheduling	Settings → **Dispatch**
Turn Razorpay on/off ("coming soon" ↔ test ↔ live), cash payments	Settings → **Payments**
Switch OpenStreetMap ↔ Google Maps, map tiles, default centre	Settings → **Maps**
Choose who gets emails/SMS, ops alert address	Settings → **Notifications**
Add/remove a service or vehicle type, edit text in English + Hindi	Catalog → **Services / Vehicle types**
Change prices	Catalog → **Rate cards** (₹ per vehicle × service: base, included km, per km, minimum)
Add a locality or an NCR city (active vs coming soon, radius)	Content → **Service areas**
Edit FAQs, testimonials	Content → **FAQs / Testimonials**
Edit Terms, Privacy, Cancellation, Grievance, partner overview	Content → **Legal pages** (Markdown, EN + HI, placeholders like `{{brand}}`)
Edit the partner agreement text (driver / fleet company)	Content → **Contract templates** (bump the version for new terms; old signed copies stay unchanged)
Approve / reject partner applications, set commission, suspend	**Partners**
Manually assign a driver / cancel / retry dispatch	**Bookings → open booking**, or the **Dispatch board**
Settle partner payouts	**Payouts & ledger** (select entries → enter UTR/reference → mark settled)
See who changed what	**Audit log**; emails/SMS sent → **Notification log**

Partner onboarding (what happens)

1. Partner applies at `/en/partner` → appears under **Partners** as *Pending*.
2. You review, set commission %, click **Approve & send agreement** → the personalised agreement is generated from the active template and emailed.

3. Partner signs in the driver app / partner portal, reads the agreement, types their name and confirms with an emailed code → account becomes **Active**.
4. Companies add their own drivers and trucks; when a booking is assigned to a company's driver, **both the driver and the company are emailed**.

Money flow cheat-sheet

- Customer pays total = taxable fare + 18% GST (editable).
- Platform commission = % of the **taxable fare** (per-partner override → company → default).
- *Cash jobs:* provider keeps the cash and owes you commission + GST collected (shown as "Providers owe platform").
- *Online jobs (Razorpay):* you hold the money and owe the provider's net (shown as "Platform owes providers").

Incident response plan (security and service incidents)

Covers cyber-security incidents (CERT-In Directions, 28 April 2022), personal-data breaches (DPDP Act, once in force) and major service outages. Review every six months and after every incident.

1. Roles

Role	Who (fill in)	Responsibility
Incident lead	Engineering lead	Runs the response, makes technical decisions
CERT-In point of contact	Founder / designated officer	Registered with CERT-In; files reports
Communications	Founder	Customers, partners, B2B clients, regulators
Scribe	Anyone on call	Timeline of actions with timestamps

Keep an up-to-date contact sheet for hosting (Vercel, Render), database (Neon), email/SMS providers, Razorpay and counsel.

2. Severity

Level	Examples	Response
SEV-1	Data breach or suspected one; admin account compromise; booking or dispatch fully down; payment fraud	Immediate; all hands; consider CERT-In report
SEV-2	Partial outage (SMS OTP failing, maps failing); a single partner account compromised	Within 30 min
SEV-3	Degraded performance; non-sensitive bug	Next business day

3. Steps

1. **Detect and record:** note the time you first noticed it. The 6-hour CERT-In clock starts here.
2. **Contain:** revoke sessions (rotate `AUTH_SECRET` signs everyone out), disable compromised staff accounts (Admin → Users), rotate leaked keys (Razorpay, SMTP, MSG91, database), switch payments to "coming soon" if payment flow is affected.
3. **Preserve evidence:** export relevant logs (hosting, database, `AuditLog`, `NotificationLog`) before they roll over. Do not wipe affected systems.
4. **Assess:** what data and how many people are affected? Is it a CERT-In specified incident type, such as unauthorised access, data breach, data leak, or attack on servers or applications?
5. **Report:**
   - **CERT-In within 6 hours** of noticing a specified incident (incident@cert-in.org.in, using the CERT-In format).
   - **Data Protection Board and affected individuals** as required by the DPDP Rules once those provisions are in force.
   - B2B clients per contract; payment provider if card or UPI data may be involved.
6. **Recover:** restore service (Neon point-in-time restore if data integrity is affected), verify, and monitor.
7. **Review within 5 working days:** root cause, what worked, actions with owners and dates. Update this plan and the risk register.

4. Logs to keep (CERT-In: rolling 180 days)

Application logs (hosting), database logs, `AuditLog`, authentication events (`Session`, `OtpChallenge`), admin actions, webhook receipts. Configure a log drain from Vercel and Render to storage under our control. The CERT-In directions speak of Indian jurisdiction, so check the current CERT-In FAQ clarifications with counsel.

5. Service-outage playbook (non-security)

Failure	Immediate workaround
Website/app down	Publish the emergency phone number on social and Google Business Profile; desk takes bookings by phone and logs them in admin when back
SMS OTP failing	Customers can use email OTP (built-in fallback); check DLT template status and provider dashboard
Maps/geocoding failing	Switch provider in Admin → Settings → Maps (OSM ⇄ Google)
Cron not running	Dispatch self-advances on tracking polls; run `/api/cron/tick` manually with the bearer secret
Database unavailable	Neon status page; failover/restore; keep phone desk running

Chapter 3 — Legal and regulatory map

Scope: what Indian law asks of a towing and roadside-assistance platform that (a) sells to consumers online, (b) works with independent tow operators and fleet companies, (c) handles personal and location data, and (d) collects money and issues GST invoices. Status is as of November 2026.

> **Not legal advice.** This is a research map to brief a Delhi startup lawyer and a chartered accountant, who must confirm every row before launch. Items marked ⚠ **confirm** are open questions where the answer depends on facts or on rules not yet notified.

3.1 Summary — what is mandatory before the first paying customer

#	Requirement	Law	Status in product	Owner
1	Incorporated entity (private limited recommended)	Companies Act 2013; Consumer Protection (E-Commerce) Rules 2020, Rule 4(1)(a)	Entity details are admin settings (placeholders today)	Founder + lawyer
2	Legal name, address, contact details and **named Grievance Officer** on the website; acknowledge complaints within 48 h, resolve within 1 month	E-Commerce Rules 2020, Rules 4 & 5; IT (Intermediary) Rules 2021	Grievance page exists; officer name field missing	Founder
3	Privacy Policy that matches real data practices	IT Act s.43A + SPDI Rules 2011; DPDP Act 2023 (phased)	Template exists; needs subprocessors, retention, cookie section	Lawyer + eng
4	Terms of Service, Cancellation & Refund policy shown **before** payment; no dark patterns	Consumer Protection Act 2019; E-Commerce Rules; Dark Patterns Guidelines 2023	Templates exist; acceptance is not recorded yet	Lawyer + eng
5	GST registration, tax invoices, returns	CGST/IGST Acts	Invoicing built (SAC, CGST/SGST/IGST split, FY numbering)	CA
6	DLT registration for SMS (sender ID + templates)	TRAI TCCCPR 2018	MSG91 adapter ready; DLT pending	Ops
7	Partner agreements (driver, fleet)	Indian Contract Act 1872; IT Act s.10A	Templates + e-sign flow built	Lawyer
8	Partner vehicle and driver compliance checks	Motor Vehicles Act 1988; CMVR 1989	Licence number field only; **document upload/verification missing**	Ops + eng
9	Security practices, 180-day logs, 6-hour incident reporting	CERT-In Directions (28 Apr 2022); SPDI Rules	Audit log exists; log-retention policy and incident plan to be written (see `docs/operations/INCIDENT-RESPONSE.md`)	Eng

3.2 Detailed map

A. Company, e-commerce and consumer protection

- **Consumer Protection (E-Commerce) Rules 2020.** These apply to entities selling goods *or services* over a digital network. Covered entities must be a Companies Act company (or a specified foreign entity). They must display legal name, address, customer-care and grievance-officer details, show the total price including taxes

before purchase, and handle complaints within the set timelines. The research you supplied notes amendments notified in September 2026 that take effect on **1 January 2027**, adding more disclosure and dark-pattern duties. ⚠ **confirm** the final amended text with counsel.

- **Guidelines for Prevention and Regulation of Dark Patterns, 2023.** No false urgency, drip pricing, forced sign-up, hidden subscription or confusing cancellation. *Product impact:* our night/surge surcharge is shown in the quote breakdown, which is good. The offer countdown is for drivers, not customers. Keep surge capped and visible.
- **Unfair contract terms.** The Consumer Protection Act 2019 lets consumer commissions strike down one-sided terms. A liability cap "limited to the fare" may not survive negligent damage to a car worth lakhs. Pair it with a real claims process and insurance (chapter 4, risk R5).

B. Information technology and data protection

- **IT Act 2000, s.43A and SPDI Rules 2011.** Covered businesses must publish a privacy policy and keep "reasonable security practices", documented in an information-security programme. ISO 27001 is one way to show this, but it is not mandatory. *Sensitive data we hold:* bank details and UPI IDs of partners, and passwords (none; we use OTPs).
- **IT (Intermediary Guidelines and Digital Media Ethics Code) Rules 2021.** Relevant to the extent the platform is an intermediary (user reviews, partner listings): publish rules and a user agreement, and name a grievance officer.
- **Digital Personal Data Protection Act 2023 and DPDP Rules 2025.** These come into force in phases. Under the commencement notification, the main notice, consent, security, breach-notification, children's-data and data-principal-rights duties apply from about **13 May 2027**. *Product impact (build before then):* itemised consent notice at booking and partner sign-up; consent records; withdrawal; **account deletion and data export** (missing today); retention schedule (e.g. booking data kept for the GST record period, analytics with no PII, as the tracking plan already requires); processor contracts with subprocessors (email, SMS, maps, Firebase, Razorpay, hosting); breach notification to the Data Protection Board and affected users.
- **Location data.** Drivers share continuous GPS while online. The partner agreement already covers this (clause 7.2). The privacy policy must say how long location history is kept. Today only the last point is stored on `Driver`; `BookingEvent` stores lat/lng at status changes.
- **CERT-In Directions 2022.** Report specified cyber incidents to CERT-In **within 6 hours** of noticing them. Keep ICT logs for a **rolling 180 days** (within Indian jurisdiction per the directions; see CERT-In FAQ on clarifications). Name a point of contact. *Product impact:* Vercel/Render logs are short-lived by default, so add a log drain to storage you control.
- **Electronic signatures. ⚠ important.** The current partner agreement says that typing a name plus an email OTP "constitutes a valid electronic signature under the IT Act 2000". Under the IT Act, a legally defined *electronic signature* means the methods notified in the Second Schedule (for example Aadhaar eSign or a digital signature certificate). A typed name with OTP is better described as an **electronically formed contract** (s.10A IT Act plus the Indian Contract Act), backed by our audit trail of time, IP and OTP. **Recommendation:** soften that sentence now, and offer Aadhaar eSign (via an ASP such as Leegality, Digio or Signzy) for fleet-company agreements.

C. Tax

- **GST registration.** Registration is compulsory above the threshold (₹20 lakh aggregate turnover for services in most states), and may be compulsory regardless of turnover for some e-commerce arrangements. ⚠ **confirm with CA.** Rate: the product defaults to **18 % under SAC 996799** (other supporting services to transport). The CA must confirm the SAC and rate per service (towing, on-spot repair and fuel delivery may differ).
- **Who supplies the customer?** This is the most important tax question:
  - *Platform as principal* (current code: platform issues the invoice on the full fare). The platform charges GST to the customer on the full value. Partners supply *to the platform*, and their own GST registration and reverse-charge questions follow. The platform deducts **TDS under s.194C** when paying partners above the thresholds.

  - *Platform as e-commerce operator (marketplace)*. Partners are the suppliers. The platform collects **GST TCS under s.52 CGST** (0.5 % CGST + 0.5 % SGST, or 1 % IGST) and deducts **income-tax TDS under s.194-O at 0.1 %** (rate from 1 Oct 2024). Towing is **not** currently a s.9(5) service, where the operator itself pays the GST.
  - The choice changes invoices, ledger and returns. The code supports the principal model today, and the ledger already separates gross, commission and net per job.
- **Cash jobs.** Partners collect the full fare including GST, and the platform recovers commission plus GST through settlements (already modelled in the ledger as "providers owe platform"). This needs a clear recovery process and credit limits (chapter 4, risk R4).

D. Transport and vehicles

- **Motor Vehicles Act 1988 and Central Motor Vehicles Rules 1989.** Tow and recovery trucks must be registered in the correct body type ("breakdown van / recovery vehicle" or "crane-mounted vehicle", per the CMVR vehicle-type notifications). They need valid fitness, permit (where applicable), insurance and PUC, and drivers need a licence of the right class. *Product impact:* the `Truck` table already has insurance, permit and fitness expiry fields. Add document upload, verification and expiry alerts before scale.
- **Motor Vehicle Aggregator Guidelines 2025 (MoRTH, 1 July 2025).** These are written for **passenger** transport aggregators: state licence, ₹5 lakh fee, driver training, AIS-140 trackers, and fare caps up to 2× base at peak. Towing is not passenger transport, so they probably do not apply directly. States adopt them through their own rules, though, and the surge-cap principle is a useful benchmark. ⚠ **confirm** with counsel and watch Delhi Transport Department notifications.
- **Accident vehicles and police.** Vehicles in accidents with an FIR or medico-legal case may need police release before removal. Operations must check before dispatching **accident recovery** (SOP in `docs/operations/OPERATIONS-SOP.md`).
- **Statutory towing.** Towing illegally parked vehicles is done by Delhi Traffic Police contractors under separate rules. It is a different market, and we must not imply any police authority.

E. Labour and gig workers

- **Code on Social Security 2020.** Came into force on **21 November 2025** together with the other labour codes. It defines gig workers, platform workers and aggregators. Aggregators will contribute **1–2 % of annual turnover, capped at 5 % of amounts paid to gig/platform workers**, to a social security fund. **The rate and start date are not yet notified.** Budget for 1 % of turnover from year 2 (chapter 6). Whether a tow-truck *business owner* (rather than a driver) counts as a "platform worker" is a ⚠ **confirm** item.
- **State gig-worker laws** (Rajasthan 2023, Karnataka 2025 and others) add registration and welfare-fee duties where we operate. Track this before expanding beyond Delhi.
- **Own employees.** EPF/ESI once headcount thresholds are crossed. An internal POSH committee once there are 10+ employees. Delhi Shops & Establishments registration for the office.

F. Payments

- **RBI Payment Aggregator directions.** Use a licensed PA (Razorpay). If the platform collects money on partners' behalf and pays them out, use the PA's split or settlement product (e.g. Razorpay Route) rather than pooling funds in our current account. ⚠ **confirm** under the final principal/marketplace model.
- **Refunds.** Commit to refund timelines in the Cancellation & Refund policy (5–7 working days today) and keep them.

G. Insurance

- Partners must carry vehicle insurance and should carry **goods-in-transit / goods-in-custody** cover for customer vehicles.
- The platform should buy **commercial general liability** and **cyber** cover once live, and consider platform-level goods-in-tow cover from year 2.

- **RSA memberships are not insurance** if sold as a prepaid service contract (call-outs included). Do not describe them as insurance or promise indemnity, which would bring them under IRDAI. ⚠ **confirm** the membership terms with counsel.

H. Intellectual property

- **Trademark search and filing** for the brand (Form TM-A, Class 39 transport/towing, Class 9/42 software). Fee: ₹4,500 per class for startups/small entities online.
- **Code ownership:** written IP assignment from every contributor (founders, employees, contractors), as required by Copyright Act s.19.
- **Open-source licences:** all dependencies are permissive (MIT/Apache/BSD/ISC). OpenStreetMap data and tiles need attribution under ODbL, which Leaflet shows by default. Public OSM tiles, Nominatim and OSRM are fair-use only, so move to a paid or self-hosted provider before real traffic.

3.3 Compliance calendar

When	Item
Before launch	Rows 1–9 of §3.1; insurance; trademark search; soften e-signature wording
Monthly / quarterly	GST returns (GSTR-1, GSTR-3B); TDS returns (quarterly); grievance register review
1 Jan 2027	E-Commerce Rules amendments take effect, so review the checkout, cancellation and pricing displays
By 13 May 2027	DPDP core obligations: consent manager readiness, deletion/export, breach process, processor contracts
On notification	Social-security fund contribution start date; any Delhi aggregator rules
Annually	ROC filings (AOC-4, MGT-7), income-tax return, statutory audit, policy review, security review

Chapter 4 — Biggest obstacles and risk register

Scoring: **Likelihood (L)** and **Impact (I)** on a 1–5 scale, **Score = L × I**. Review monthly; anything at 15 or above goes on the management agenda.

4.1 The five obstacles that decide success

1. **Supply density, 24×7.** A customer stuck at 2 a.m. in Dwarka needs a truck *near Dwarka at 2 a.m.* Arrival time is the product. With too few partners online, the dispatcher sees `NO_DRIVER_FOUND`, and one bad night can lose a customer for life. *Answer:* launch in a limited set of zones, guarantee minimum earnings for night partners in the pilot, and expand only when median arrival stays under 35 minutes.
2. **Customer acquisition cost vs. low frequency.** People need a tow rarely, perhaps once every few years. Paid search for "towing near me Delhi" is expensive, and our contribution is about ₹210 per job (chapter 6). *Answer:* SEO (locality pages are already built), partnerships with dealers, insurers, fleets, parking operators and housing societies, and memberships that turn a one-off into a recurring customer.
3. **Leakage (disintermediation).** After one job, the customer has the driver's number and may call directly next time. *Answer:* masked calling, platform-only benefits (invoice, damage cover, tracking, membership), a partner agreement clause against solicitation (already in clause 3.4), and making the platform the partner's best source of jobs so leaving costs them more than staying.
4. **Trust and damage claims.** One scratched luxury car or one rude driver can go viral. *Answer:* KYC'd partners, trip PIN (built), photo condition report at pickup and drop (to build, AI-assisted in chapter 5), mandatory partner insurance, a fast claims SOP, and ratings with automatic suspension.
5. **Incumbent B2B lock-in.** Most new cars and insured cars already come with "free" RSA from Allianz, Europ, TVS Auto Assist and others. *Answer:* sell to these companies as their Delhi fulfilment partner instead of fighting them, and win uncovered customers: older vehicles, two-wheelers, commercial vehicles, and people whose RSA failed them.

4.2 Risk register

ID	Risk	Category	L	I	Score	Mitigation	Owner	Early-warning signal
R1	Not enough partners online in a zone or hour; long arrivals	Operations	4	5	**20**	Zone-limited launch; night incentives; fleet-company partners; demand heatmaps (AI-4)	Ops lead	Time-to-assign > 5 min; `NO_DRIVER_FOUND` > 5 %
R2	CAC higher than contribution per job	Commercial	4	4	**16**	SEO, B2B channels, memberships, referral credits; stop any paid channel with CAC > 1.5× contribution	Founder	Blended CAC > ₹300
R3	Customers or partners transact off-platform	Commercial	4	4	**16**	Masked calls, platform-only value, anti-solicitation clause, quality bonuses	Founder	Repeat rate falling while partner job count steady
R4	Partners don't remit commission and GST on cash jobs	Financial	3	4	12	Credit limit per partner; offset against online earnings; auto-pause above a balance; push online payment	Finance	"Providers owe platform" balance > 14 days
R5	Customer vehicle damaged; claim exceeds partner's means	Legal/Trust	3	5	**15**	Partner insurance mandatory; photo evidence; claims SOP; platform cover from year 2	Ops lead	Any claim > ₹25,000

ID	Risk	Category	L	I	Score	Mitigation	Owner	Early-warning signal
R6	Safety incident (assault, theft, accident during tow)	Safety	2	5	10	KYC and police verification of drivers, trip PIN, live tracking, SOS contact, suspension policy	Ops lead	Any incident
R7	Regulatory change: aggregator licensing, social-security contribution, DPDP, e-commerce amendments	Legal	3	3	9	Compliance calendar (chapter 3); budget 1 % of turnover for social security; counsel on retainer	Founder	Gazette notifications
R8	Wrong GST/TDS/TCS model causes back taxes and penalties	Tax	3	4	12	Decide principal vs. marketplace with CA before launch; monthly reconciliations	Finance + CA	CA review findings
R9	Personal-data breach	Security	2	5	10	Security baseline (`docs/developer/SECURITY.md`), CERT-In incident plan, least-privilege admin roles, log retention	Eng lead	Unusual admin logins; rate-limit spikes
R10	OSM public endpoints throttle us (geocoding/routing)	Technology	4	3	12	Self-host Nominatim/OSRM, or switch to Google/Mappls (provider interface already exists)	Eng lead	429 errors from geo API
R11	SMS OTP delivery failures (DLT, operator filtering)	Technology	3	4	12	DLT registration; Firebase phone auth with email fallback (built); WhatsApp OTP later	Eng lead	OTP verify rate < 85 %
R12	Polling load grows with live tracking at scale	Technology	2	3	6	SSE/WebSockets + Redis for locations (scaling path in ARCHITECTURE.md)	Eng lead	DB CPU > 60 % sustained
R13	Fraud: fake bookings, GPS spoofing, collusive fake jobs to earn incentives	Fraud	3	3	9	OTP-verified accounts; PIN-gated start; GPS sanity checks; anomaly detection (AI-7)	Eng + Ops	Completed jobs with zero movement
R14	Key-person dependency (solo founder, single developer)	Organisation	4	4	**16**	Documentation (this pack), second engineer, shared credentials vault, runbooks	Founder	—
R15	Funding gap before break-even	Financial	3	5	**15**	Lean pilot first; 18–24 months runway at seed; monthly burn review	Founder	Runway < 9 months
R16	Brand/trademark conflict on "RoadSaathi"	Legal	2	3	6	Trademark search before marketing spend; brand is a single setting to change	Founder	Search result
R17	Accounting or governance failure (GoMechanic case)	Governance	1	5	5	Ledger is source of truth; monthly close; external audit; no booking of unrealised revenue	Finance	Ledger vs. bank mismatch
R18	24×7 desk burnout and attrition	People	3	3	9	Shift rota, automation of routine steps, clear escalation, fair pay	Ops lead	Missed SLAs at night

ID	Risk	Category	L	I	Score	Mitigation	Owner	Early-warning signal
R19	Weather surges (monsoon waterlogging, fog season) overwhelm supply	Operations	4	3	12	Seasonal partner standby agreements; surge cap ≤ 1.5× with clear display; pre-positioning	Ops lead	IMD alerts
R20	A big B2B client takes most of the volume (concentration, as at Urgently)	Commercial	2	4	8	No single client > 30 % of revenue; keep the consumer channel	Founder	Concentration report

4.3 Top risks heat view

Impact ↓ / Likelihood →	1	2	3	4	5
5	R17	R6, R9	R5, R15	R1	
4		R20	R4, R8, R11	R2, R3, R14	
3		R12, R16	R7, R13, R18	R10, R19	